



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
17-10-2021

JEE(Main + Advanced) : NURTURE COURSE [PHASE : TEPS, TOPS, I(A), I, I(B), II(A), II & LIVE-I, I(B), II(A)]

Time : 3 Hours

PAPER-2

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. A particle is projected from a tower of height (H) with a speed of 50 m/s at an angle of 37° with the horizontal. Then select the **CORRECT** alternative(s) (take point of projection as origin) :

(A) Equation of trajectory of particle is $y = \frac{3}{4}x - \frac{x^2}{320}$.

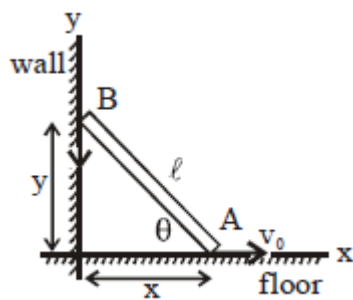
(B) Particle moves perpendicular to initial direction at $\frac{25}{3}$ sec.

(C) Minimum speed of particle in subsequent motion is zero.

(D) Maximum height attained by the particle from the point of projection is 45 m.

Space for Rough Work

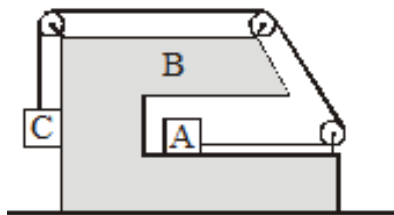
2. A rigid rod of length ℓ is sliding. At some instant, position of the rod is as shown in the figure. End A has constant velocity v_0 . At $t = 0$, $y = \ell$.



- (A) Component of velocity of end B along the length of the rod as a function of time t is $\frac{v_0^2 t}{\ell}$
- (B) Graph of 'y' as a function of time t is a parabola.
- (C) Velocity of end B as a function of time t is $\frac{v_0^2 t}{y}$
- (D) Graph of 'y' as a function of time t is an ellipse

Space for Rough Work

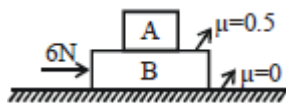
3. For given figure m_A is 30 kg, $m_B = m_C = 5$ kg respectively. All contact surfaces are smooth. ($g = 10 \text{ m/s}^2$). Select the correct statements.



- (A) Acceleration of block A, B and C are 1 m/s^2 , 4 m/s^2 and 5 m/s^2 respectively
- (B) Contact force between B & Ground is 350 N
- (C) Tension in string is 30N
- (D) Contact force between B and C is 20 N
4. A particle is moving with constant speed u inside a fixed smooth hemispherical bowl of radius a describes a horizontal circle at a distance $\frac{\sqrt{3}a}{2}$ below its centre. Then
- (A) The radius of the circular motion is $\frac{a}{2}$
- (B) The value of u is $\sqrt{\frac{ag}{2\sqrt{3}}}$
- (C) The normal reaction of the spherical surface on the particle is $\frac{mg}{2}$
- (D) The magnitude of the resultant force acting on the particle is zero, in an inertial frame.

Space for Rough Work

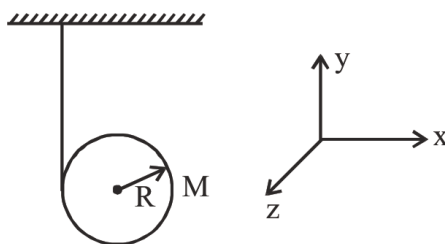
5. Block A of mass 1 kg is kept on block B of mass 2 kg which is kept on smooth surface as shown in figure. At $t = 0$ both blocks are at rest & a force of 6N is applied on block B. Friction coefficient between A & B is 0.5. Which of the following is/are correct :-



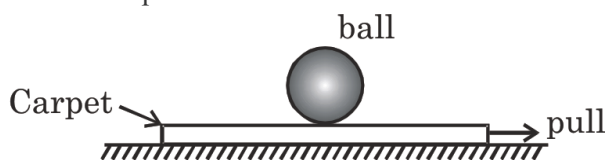
- (A) Acceleration of both the blocks is 2 m/s^2
- (B) Work done by friction (in ground frame) on A is zero from $t = 0$ to $t = 2 \text{ sec}$
- (C) Friction force on A is 2N
- (D) Kinetic energy of A at $t = 2 \text{ sec}$ is 8J
6. If resultant of two vectors $\vec{A}$ and $\vec{B}$ is perpendicular to $\vec{B}$. If $|\vec{A}| = 4\sqrt{3}$, $|\vec{B}| = 6$ then :
- (A) Angle between $\vec{A}$ and $\vec{B}$ is $\frac{\pi}{6}$ radian.
- (B) Magnitude of resultant is $2\sqrt{3}$.
- (C) $\vec{B} \cdot (\vec{A} + \vec{B}) = 0$
- (D) Angle between $(\vec{A} + \vec{B})$ and $\vec{A}$ is acute angle.

Space for Rough Work

7. A disc of mass M radius R moves under gravity while unwinding string under non slip condition. Mark the correct statement(s):



- (A) Acceleration of the centre of the disc is in $-\hat{j}$ direction
- (B) Rate of change of angular momentum $\left(\frac{d\vec{L}}{dt}\right)$ of disc $\frac{d\vec{L}}{dt}$ about a point along string is in $-\hat{k}$ direction.
- (C) Angular momentum of disc is conserved about centre of mass.
- (D) Rate of change of angular momentum $\frac{d\vec{L}}{dt}$ of disc about center of mass is in $+\hat{k}$ direction.
8. A ball lies on a carpet. Carpet is kept on smooth horizontal floor. Carpet is pulled as shown in figure. There is no slipping between ball & carpet :



- (A) Acceleration of center of mass of ball is in the direction of acceleration of carpet.
- (B) Acceleration of center of mass of ball is in opposite direction of acceleration of carpet.
- (C) Direction of angular acceleration of ball is out of the plane of paper.
- (D) Work done by friction on ball + carpet system is non-zero.

Space for Rough Work

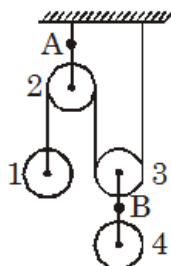
SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **Two (02)** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

In the figure shown, the masses of each of the pulley disc is m . They were at rest. All the threads are light and inextensible. Now the thread is cut at B. Denoting the acceleration of the discs as a_1, a_2, a_3, a_4 respectively, match the correct value of acceleration thereafter



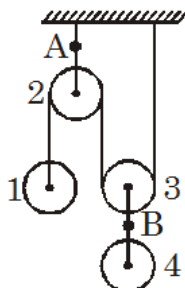
List-I		List-II	
(I)	a_1	(P)	0 m/s^2
(II)	a_2	(Q)	10 m/s^2
(III)	a_3	(R)	$80/9 \text{ m/s}^2$
(IV)	a_4	(S)	$110/9 \text{ m/s}^2$
		(T)	4 m/s^2
		(U)	2 m/s^2

- (A) I $\rightarrow$ U; II $\rightarrow$ P; III $\rightarrow$ T; IV $\rightarrow$ Q
 (B) I $\rightarrow$ T; II $\rightarrow$ P; III $\rightarrow$ T; IV $\rightarrow$ Q
 (C) I $\rightarrow$ U; II $\rightarrow$ P; III $\rightarrow$ U; IV $\rightarrow$ Q
 (D) I $\rightarrow$ T; II $\rightarrow$ P; III $\rightarrow$ U; IV $\rightarrow$ Q

Space for Rough Work

10. Answer the following by appropriately matching the lists based on the information given in the paragraph.

In the figure shown, the masses of each of the pulley disc is m . They were at rest. All the threads are light and inextensible. Now the thread is cut at A. Denoting the acceleration of the discs as a_1, a_2, a_3, a_4 respectively, match the correct value of acceleration thereafter



List-I		List-II	
(I)	a_1	(P)	0 m/s^2
(II)	a_2	(Q)	10 m/s^2
(III)	a_3	(R)	$60/7 \text{ m/s}^2$
(IV)	a_4	(S)	$90/7 \text{ m/s}^2$
		(T)	4 m/s^2
		(U)	2 m/s^2

(A) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ T; IV $\rightarrow$ T

(B) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ R; IV $\rightarrow$ R

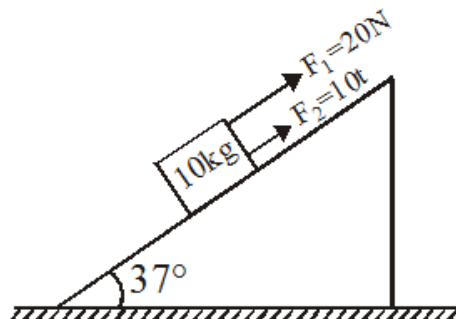
(C) I $\rightarrow$ U; II $\rightarrow$ T; III $\rightarrow$ U; IV $\rightarrow$ U

(D) I $\rightarrow$ R; II $\rightarrow$ Q; III $\rightarrow$ R; IV $\rightarrow$ R

Space for Rough Work

11. Answer the following by appropriately matching the lists based on the information given in the paragraph.

A block of mass 10 kg is at rest on rough inclined of inclination 37° , with help of a constant force F_1 . Force F_1 have magnitude 20 N. At $t = 0$ another force F_2 start acting on block. Magnitude of force F_2 changes with time as $F_2 = 10t$ N. Both forces F_1 and F_2 are parallel to inclined and are acting up the inclined. Friction force acting on block is f . Match the list-I and List-II.



Coefficient of friction between block and inclined is 1.

	List-I		List-II
(I)	$t = 4$ sec.	(P)	$f = 0$
(II)	$t = 7$ sec	(Q)	$f = 30$ N
(III)	$t = 10$ sec.	(R)	$f = 40$ N
(IV)	$t = 15$ sec.	(S)	$f = 60$ N
		(T)	$f = 80$ N
		(U)	$f = 100$ N

(A) I $\rightarrow$ S; II $\rightarrow$ T; III $\rightarrow$ Q; IV $\rightarrow$ R

(B) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ R; IV $\rightarrow$ R

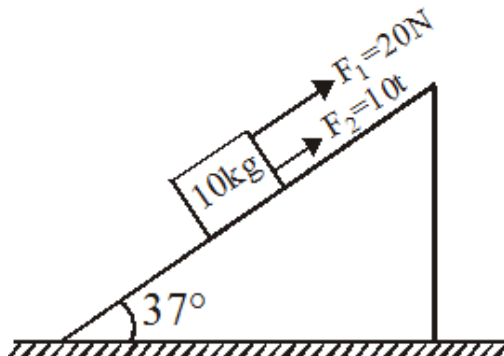
(C) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ S; IV $\rightarrow$ T

(D) I $\rightarrow$ S; II $\rightarrow$ Q; III $\rightarrow$ R; IV $\rightarrow$ P

Space for Rough Work

12. Answer the following by appropriately matching the lists based on the information given in the paragraph.

A block of mass 10 kg is at rest on rough inclined of inclination 37° , with help of a constant force F_1 . Force F_1 have magnitude 20 N. At $t = 0$ another force F_2 start acting on block. Magnitude of force F_2 changes with time as $F_2 = 10t$ N. Both forces F_1 and F_2 are parallel to inclined and are acting up the inclined. Friction force acting on block is f . Match the list-I and List-II



Coefficient of friction between block and inclined is 0.5.

	List-I		List-II
(I)	$t = 4$ sec.	(P)	$f = 0$
(II)	$t = 7$ sec	(Q)	$f = 30$ N
(III)	$t = 10$ sec.	(R)	$f = 40$ N
(IV)	$t = 15$ sec.	(S)	$f = 60$ N
		(T)	$f = 80$ N
		(U)	$f = 100$ N

(A) $I \rightarrow P; II \rightarrow Q; III \rightarrow R; IV \rightarrow P$

(B) $I \rightarrow P; II \rightarrow Q; III \rightarrow R; IV \rightarrow R$

(C) $I \rightarrow R; II \rightarrow P; III \rightarrow Q; IV \rightarrow Q$

(D) $I \rightarrow Q; II \rightarrow Q; III \rightarrow P; IV \rightarrow P$

Space for Rough Work

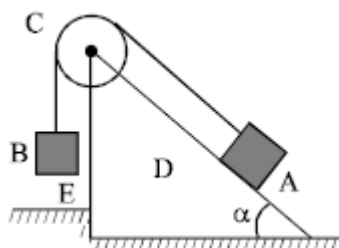
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

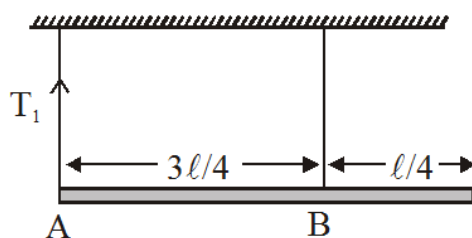
Zero Marks : 0 In all other cases.

1. A particle of mass 'm' and velocity ' $\vec{v}$ ' collides oblique elastically with a stationary particle of mass 'm'. The angle between the velocity vectors of the two particles after the collision is (in degree):
2. A body A weighing W_1 descends down on an inclined plane D, which makes an angle α with the horizontal, and pulls a load B that weighs W_2 by means of a weightless and inextensible thread passing over a small light pulley C, as shown in figure. If the horizontal component of the force is $\frac{2K}{75}$ (in newton) which the inclined plane D applies on the floor corner E. Find the value of K. (Take $\alpha = 37^\circ$, $W_1 = 40$ N, $W_2 = 20$ N, neglect friction, $g = 10$ m/s²)

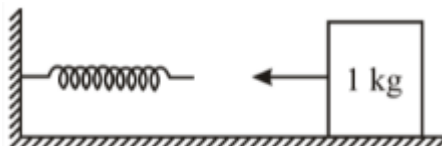


Space for Rough Work

3. A uniform rod of mass 12 kg and length 4 m is hung from two string of equal length from a ceiling as shown in fig. Determine the tension T_1 (in N) in the string. (take $g = 10 \text{ m/s}^2$)



4. For motion of an object along x-axis, the velocity v depends on the displacement x as $v = 3x^2 - 2x$. What is the acceleration at $x = 2\text{m}$ (in m/s^2) ?
5. A body of mass 1 kg is thrown upwards with a velocity 20 m/s. It momentarily comes to rest after attaining a height of 18m. How much energy is lost due to air friction ? (In Joule) ($g = 10 \text{ m/s}^2$)
6. 1 kg block collides with a horizontal massless spring of force constant 2 N/m. The block compresses the spring by 4m. The coefficient of kinetic friction between the block and the surface is 0.25. If the speed of the block at the instant of collision is $\sqrt{x}$ then the value of x is ? (take $g = 10 \text{ m/s}^2$)



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

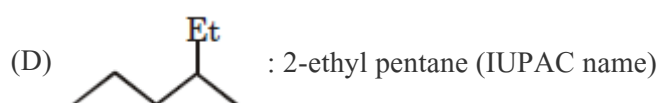
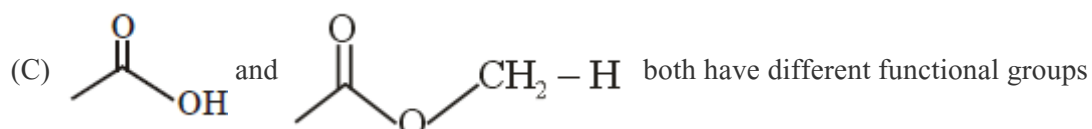
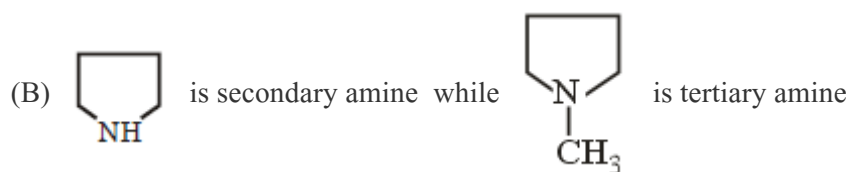
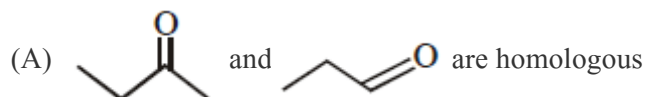
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Neohexane contains :

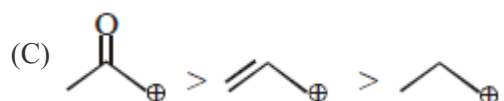
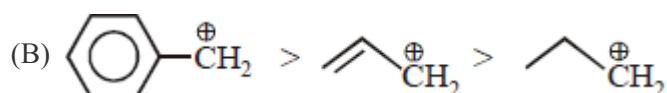
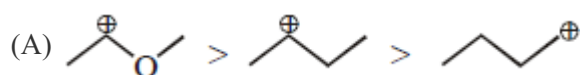
- (A) Four 1° carbon atoms and two 2° hydrogen
- (B) Twelve 1° hydrogen but no 3° carbon
- (C) Two 2° carbon and one 4° carbon
- (D) One 4° carbon and two 3° hydrogen

Space for Rough Work

2. Which of the following is/are correct :



3. Correct order of carbocation stability is/are :



Space for Rough Work

4. 25 ml 0.4M- Na_3PO_4 solution is mixed with 50 ml 0.2M- BaCl_2 solution and the resulting solution is diluted to 200 ml. Assuming that $\text{Ba}_3(\text{PO}_4)_2$ is completely insoluble in water, the correct molar concentrations of ions in the final solution is/are -
- (A) $[\text{Ba}^{2+}] = 0$ (B) $[\text{PO}_4^{3-}] = 0$ (C) $[\text{Na}^+] = 0.15\text{M}$ (D) $[\text{Cl}^-] = 0.1\text{M}$
5. A solution is prepared by dissolving a solid mixture of $\text{K}_2\text{C}_2\text{O}_4$ and KHC_2O_4 . A 10 ml portion of this solution required 10 ml, 0.05 M KOH solution for complete titration. In a separate analysis 10 ml of the same stock solution required 10 ml, 0.06 M acidified KMnO_4 solution for complete titration. Which of the following is/are CORRECT :
- (A) The original mixture contains $\text{K}_2\text{C}_2\text{O}_4$ and KHC_2O_4 in 2 : 1 molar ratio
- (B) 20 ml of the original stock solution require 3 millimole of acidified dichromate solution for titration
- (C) 20 ml of the original stock solution requires 6 milli-equivalents of acidified dichromate solution for complete titration
- (D) All of these
6. In which of the following case increase in Z_{eff} for valence electron is/are same (as per Slater's rule)?
- (A) $\text{Na} \rightarrow \text{Mg}$ (B) $\text{C} \rightarrow \text{N}$ (C) $\text{Ti} \rightarrow \text{V}$ (D) $\text{Rb} \rightarrow \text{Cs}$
7. Molecules in which any hybrid orbital(s) of central atom which form bond, contains s-character less than 25%.
- (A) PH_3 (B) H_2O (C) NH_3 (D) H_2S
8. Which statement is/are correct about NH_2^- , NH_3 , NH_4^+ :
- (A) Hybridization of N is same (B) Lewis basic strength is same
- (C) Electronic geometry is same (D) $\text{NH}_2^- > \text{NH}_3 > \text{NH}_4^+$: Bond length (N-H)

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **Two (02)** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

9. Answer the following by appropriately matching the lists based on the information given in the paragraph.

	List - I Reaction)		List - II Equivalent mass of underlined molecule in terms of molar mass M)
(I)	$\text{Zn} + \underline{\text{HNO}_3} \rightarrow \text{Zn}(\text{NO}_3)_2 + \text{NH}_4\text{NO}_3 + \text{H}_2\text{O}$	(P)	$\frac{M}{1}$
(II)	$\text{H}_2\text{O}_2 \rightarrow \underline{\text{H}_2\text{O}} + \text{O}_2$	(Q)	$\frac{M}{2}$
(III)	$\left(\underline{\text{NH}_4}\right)_2\text{Cr}_2\text{O}_7 \rightarrow \text{N}_2 + \text{Cr}_2\text{O}_3 + \text{H}_2\text{O}$	(R)	$\frac{5M}{4}$
(IV)	$\underline{\text{P}_4} + \text{NaOH} + \text{H}_2\text{O} \rightarrow \text{PH}_3 + \text{NaH}_2\text{PO}_2$	(S)	$\frac{M}{6}$
		(T)	$\frac{M}{3}$
		(U)	$\frac{3M}{2}$

Which of the following has the correct combination considering **List-I** and **List-II** ?

- (A) II $\rightarrow$ Q (B) I $\rightarrow$ P (C) III $\rightarrow$ T (D) IV $\rightarrow$ T

Space for Rough Work

10. Answer the following by appropriately matching the lists based on the information given in the paragraph.

	List - I Reaction)		List - II (Equivalent mass of underlined molecule in terms of molar mass M)
(I)	$\text{Zn} + \underline{\text{HNO}_3} \rightarrow \text{Zn}(\text{NO}_3)_2 + \text{NH}_4\text{NO}_3 + \text{H}_2\text{O}$	(P)	$\frac{M}{1}$
(II)	$\text{H}_2\text{O}_2 \rightarrow \underline{\text{H}_2\text{O}} + \text{O}_2$	(Q)	$\frac{M}{2}$
(III)	$\left(\underline{\text{NH}_4}\right)_2\text{Cr}_2\text{O}_7 \rightarrow \text{N}_2 + \text{Cr}_2\text{O}_3 + \text{H}_2\text{O}$	(R)	$\frac{5M}{4}$
(IV)	$\underline{\text{P}_4} + \text{NaOH} + \text{H}_2\text{O} \rightarrow \text{PH}_3 + \text{NaH}_2\text{PO}_2$	(S)	$\frac{M}{6}$
		(T)	$\frac{M}{3}$
		(U)	$\frac{3M}{2}$

Which of the following has the incorrect combination considering **List-I** and **List-II** ?

- (A) I $\rightarrow$ R (B) II $\rightarrow$ P (C) III $\rightarrow$ R (D) IV $\rightarrow$ T

Space for Rough Work

11. Answer the following by appropriately matching the lists based on the information given in the paragraph.
In the following List-I contains some pairs of atoms and List-II contains order for their different properties.

List-I		List-II	
(I)	NH_3	(P)	$\text{p}\pi$ - $\text{p}\pi$ bond is present
(II)	POCl_3	(Q)	lone pair on central atom is one
(III)	SOCl_2	(R)	Bond angle = 120°
(IV)	COCl_2	(S)	$120^\circ > \text{bond angle} > 109^\circ 28'$
		(T)	Tetrahedral shape
		(U)	Involves d-orbital in bonding

Which of the following options have **INCORRECT** combination considering List(I) and List(II)?

- (A) $\text{I} \rightarrow \text{Q}$ (B) $\text{II} \rightarrow \text{U}$ (C) $\text{III} \rightarrow \text{T}$ (D) $\text{IV} \rightarrow \text{P}$
12. Answer the following by appropriately matching the lists based on the information given in the paragraph.
In the following List-I contains some pairs of atoms and List-II contains order for their different properties.

List-I		List-II	
(I)	NH_3	(P)	$\text{p}\pi$ - $\text{p}\pi$ bond is present
(II)	POCl_3	(Q)	lone pair on central atom is one
(III)	SOCl_2	(R)	Bond angle = 120°
(IV)	COCl_2	(S)	$120^\circ > \text{bond angle} > 109^\circ 28'$
		(T)	Tetrahedral shape
		(U)	Involves d-orbital in bonding

Which of the following option have **CORRECT** combination considering List(I) and List(II)?

- (A) $\text{I} \rightarrow \text{S}$ (B) $\text{II} \rightarrow \text{R}$ (C) $\text{III} \rightarrow \text{P}$ (D) $\text{IV} \rightarrow \text{S}$

Space for Rough Work

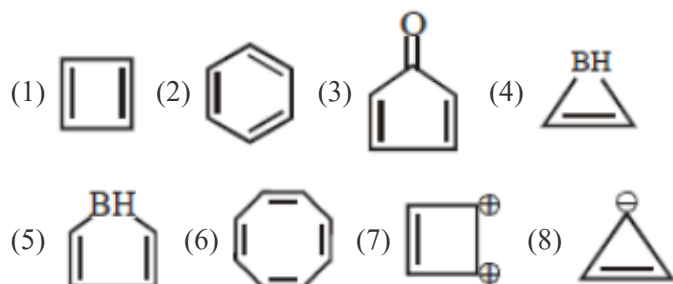
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. How many compound are unstable at room temperature.



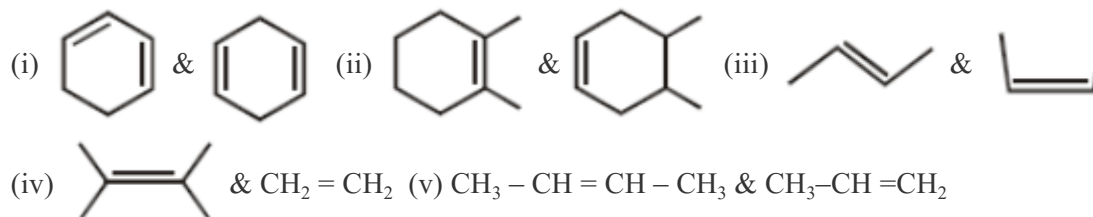
2. How many of these group shows -R effect when attached to benzene :

(1) $-\text{NO}_2$ (2) $-\text{CH}_3$ (3) $-\text{OCH}_3$ (4) $-\text{COCH}_3$ (5) $-\text{NH}_2$

(6) $-\text{CONH}_2$ (7) $-\text{CHO}$ (8) $-\text{CN}$ (9) $-\text{COCl}$

Space for Rough Work

3. In how many pairs of compounds have same order of heat of hydrogenation an heat of combustion.



4. How many blood cells of volume 5 ml each, having $[\text{K}^+] = 0.1 \text{ M}$ should burst into 25 ml of blood plasma having initial $[\text{K}^+] = 0.02 \text{ M}$ so as to give final $[\text{K}^+] = 0.06 \text{ M}$
5. Calculate the mass (in gm) of urea (NH_2CONH_2) containing 1 gm-atom H.
6. How many of the following orders are correct with respect to the indicated properties?
- (i) $\text{HCl} < \text{HF}$ Boiling point
 - (ii) $\text{PH}_3 < \text{NH}_3$ Dipole moment
 - (iii) $\text{OH}_2 < \text{D}_2\text{O}$ Strength of Hydrogen bonding
 - (iv) $\text{F}_2 < \text{Cl}_2$ Bond energy
 - (v) $\text{H}_2\text{O} < \text{OCl}_2$ Bond angle
 - (vi) $\text{H}_2\text{O}_2 < \text{O}_2\text{F}_2$ Bond length of O—O bond
 - (vii) $\text{Cr}_2\text{O}_3 > \text{CrO}_3$ Basic strength
 - (viii) $\text{SO}_2 < \text{SO}_3$ Acidic strength

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. If the equation $ax^2 - 6xy + y^2 + bx + cy + d = 0$ represents pair of lines whose slopes are m and m^2 , then value of a is/are -
 (A) $a = -8$ (B) $a = 8$ (C) $a = 27$ (D) $a = -27$
2. The lines $x + y - 1 = 0$, $(m - 1)x + (m^2 - 7)y - 5 = 0$ and $(m - 2)x + (2m - 5)y = 0$ are -
 (A) concurrent for three values of m (B) concurrent for no value of m
 (C) parallel for one value of m (D) parallel for two values of m

Space for Rough Work

3. If system of equation $px + qy + rz = 0$, $qx + ry + pz = 0$ & $rx + py + qz = 0$ has a unique solution (here $p, q, r \in \mathbb{R}$ & $p \neq q \neq r$), then system of equations
- $$(q + r)x + (r + p)y + (p + q)z = 0,$$
- $$(r + p)x + (p + q)y + (r + q)z = 0,$$
- $$(p + q)x + (q + r)y + (r + p)z = 0$$
- (A) has only one solution (B) has infinite many solution
(C) has no solution (D) is consistent
4. If 4 distinct positive numbers : $a \log_c 2, b^3, \frac{c}{b^3}, \frac{64}{\log_c 2^a}$ are in H.P, then 'c' can't be -
- (A) 1 (B) 32 (C) 64 (D) 128
5. If $\tan^3 x - 5 \tan^2 x + 5 \tan x - 1 = 0$, then which of the following is(are) true ?
- (A) There will be 3 solutions in $(0, \pi)$ (B) There will be 4 solutions in $(0, 2\pi)$
(C) There will be 2 solutions in $(0, \pi)$ (D) There will be 6 solutions in $(0, 2\pi)$

Space for Rough Work

6. The value's of 'x' satisfying $\tan^2 x - 2\sqrt{2} \tan x + 1 \leq 0$ is-
- (A) $\left[\frac{\pi}{8}, \frac{3\pi}{8}\right]$ (B) $\left[\frac{9\pi}{8}, \frac{11\pi}{8}\right]$ (C) $\left[-\frac{7\pi}{8}, -\frac{5\pi}{8}\right]$ (D) $\left[-\frac{15\pi}{8}, -\frac{13\pi}{8}\right]$
7. In ΔABC (with usual notations) if, $\sin A : \sin B : \sin C = 7 : 25 : 24$. Then which of the following can be converted into unity-
- (A) $2\log_a(c^2 - b^2)$ (B) $\log_a \sqrt{b^2 - c^2}$
- (C) $(ab)^{\log \frac{a}{b}} \times (bc)^{\log \frac{b}{c}} \times (ca)^{\log \frac{c}{a}}$ (D) $\left(\frac{a}{b}\right)^{\log ab} \times \left(\frac{b}{c}\right)^{\log bc} \times \left(\frac{c}{a}\right)^{\log ca}$
8. An equation of a line through the point (3,1) whose distance from point (4,2) is greatest, is-
- (A) $y + 2 = x$ (B) $y = x + 2$ (C) $x + y = 4$ (D) $2y = x - 1$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **Two (02)** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

9. If in a ΔABC , $a^2 + 4b^2 - 4ab = 2ac - a^2 - c^2$, then match the following :

List-I		List-II	
(I)	$\cos A$	(P)	$\frac{1}{4}$
(II)	$\cos B$	(Q)	$\frac{7}{8}$
(III)	$\cos C$	(R)	0
(IV)	$\sin(A - C)$	(S)	1
		(T)	$-\frac{1}{2}$
		(U)	$\frac{3}{4}$

Which of the following is the only **CORRECT** combination ?

- (A) $I \rightarrow S$ (B) $II \rightarrow Q$ (C) $I \rightarrow T$ (D) $III \rightarrow Q$

Space for Rough Work

10. If in a ΔABC , $a^2 + 4b^2 - 4ab = 2ac - a^2 - c^2$, then match the following :

List-I		List-II	
(I)	$\cos A$	(P)	$\frac{1}{4}$
(II)	$\cos B$	(Q)	$\frac{7}{8}$
(III)	$\cos C$	(R)	0
(IV)	$\sin(A - C)$	(S)	1
		(T)	$-\frac{1}{2}$
		(U)	$\frac{3}{4}$

Which of the following is the only INCORRECT combination ?

- (A) III $\rightarrow$ P (B) IV $\rightarrow$ R (C) II $\rightarrow$ Q (D) III $\rightarrow$ Q

Space for Rough Work

11.

List-I		List-II	
(I)	Number of real roots of $e^x + e^{2x} - 5 = 0$ is	(P)	0
(II)	Sum of infinite positive terms of G.P. is 4 and common ratio r and sum of square of these terms is 8 then value of $\frac{1}{r}$ is	(Q)	1
(III)	Number of real roots of $x^4 + \sqrt{x^4 + 10} = 10$ is	(R)	2
(IV)	Number of integral solutions of equation $\log_2(1 + \sqrt{7x - x^2 - 6}) > 0$ is	(S)	3
		(T)	4
		(U)	5

Which of the following is the only CORRECT combination ?

- (A) $I \rightarrow S$ (B) $II \rightarrow Q$ (C) $I \rightarrow Q$ (D) $III \rightarrow Q$

Space for Rough Work

12.

List-I		List-II	
(I)	Number of real roots of $e^x + e^{2x} - 5 = 0$ is	(P)	0
(II)	Sum of infinite positive terms of G.P. is 4 and common ratio r and sum of square of these terms is 8 then value of $\frac{1}{r}$ is	(Q)	1
(III)	Number of real roots of $x^4 + \sqrt{x^4 + 10} = 10$ is	(R)	2
(IV)	Number of integral solutions of equation $\log_2(1 + \sqrt{7x - x^2 - 6}) > 0$ is	(S)	3
		(T)	4
		(U)	5

Which of the following is the only CORRECT combination ?

- (A) $I \rightarrow S$ (B) $II \rightarrow S$ (C) $IV \rightarrow U$ (D) $III \rightarrow Q$

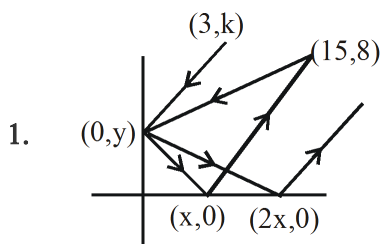
Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.



As shown in the figure, a ray of light leaves the point $(3, k)$ reflects off the y -axis towards the x -axis and reflects off the x -axis to reach the point $(15, 8)$, now a ray of light leaves the points $(15, 8)$ reflects off the y -axis towards the x -axis and reflects off the x -axis then value of $\frac{5k}{3}$ is

2. If $\sqrt{3}$, H , $\sqrt{2}$ are in H.P., then the value of $\frac{H + \sqrt{2}}{H - \sqrt{2}} + \frac{H + \sqrt{3}}{H - \sqrt{3}}$ is

Space for Rough Work

3. If α be the common root of the equation $x^3 + ax + 2 = 0$ and $x^4 + ax^2 + 6 = 0$, then $|\alpha + 15\alpha|$ has value
4. If $(0, \alpha)$ lies inside the triangle formed by the lines $y + 3x + 2 = 0$, $3y - 2x - 5 = 0$ and $4y + x - 14 = 0$ then sum of all the integral values of α is
5. The value of $\left| \begin{array}{cc} \log_3 512 & \log_4 3 \\ \log_3 8 & \log_4 9 \end{array} \right| \left| \begin{array}{cc} \log_2 3 & \log_8 3 \\ \log_3 4 & \log_3 4 \end{array} \right|$ is
6. Let $P(\alpha, \beta)$ be a point on the line $x = 1$, such that $PA - PB$ is maximum, where $A \equiv (6, 0)$ & $B \equiv (0, 3)$, then $\alpha + \beta$ is equal to -

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work